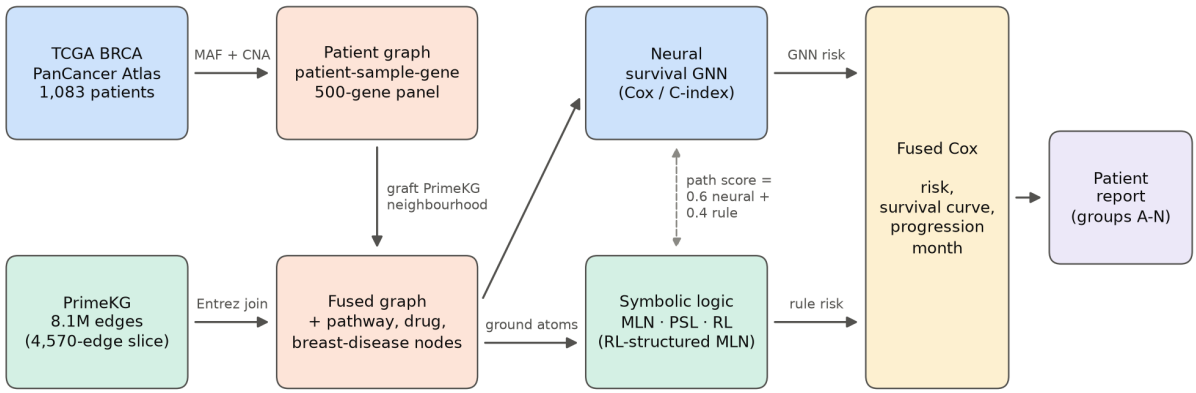
Project workflow at a glance

Neuro-symbolic reasoning over patient knowledge graphs for breast cancer progression

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What the project does

For each of 1,083 breast cancer patients (TCGA PanCancer Atlas 2018, 145 recorded progressions), we build a knowledge graph linking the patient to their altered genes and to PrimeKG pathways, drugs, and diseases. The system then estimates the risk of progression and its likely month, explains the estimate with knowledge-graph paths and weighted logic formulas, and writes a PDF report for every patient.



The flow of work, stage by stage

Stage	What happens	Code
1. Data	Clinical, mutation, and copy-number tables from cBioPortal, checked against official checksums. Outcome columns are blocked from every input.	`src/clinical.py, src/labels.py`
2. Patient graphs	Patient → sample → altered gene, limited to a 500-gene panel with 27 known drivers.	`scripts/build_patient_graphs.py`
3. PrimeKG fusion	Genes join PrimeKG on Entrez ID. A 4,570-edge slice adds partners, pathways, drugs, and breast diseases (2 hops).	`src/dataset_fusion.py`
4. Neural	Seven GNNs tested. GraphSAGE with a C-index loss gives the neural risk, and GNNExplainer scores every edge.	`src/gnn_models.py`
5. Symbolic	422 facts per patient. Markov logic, soft logic, and RL rules compared against RuleKit. The RL-structured survival MLN is the engine.	`src/neurosymbolic/logic/`
6. Fusion	One Cox model on clinical fields, the GNN risk, and the logic risk. Chosen by the one-standard-error rule.	`scripts/run_neurosymbolic_pipeline.py`
7. Month	A log-logistic time model, fitted on past progressors, estimates the month of progression.	`src/neurosymbolic/timing.py`
8. Paths	Each path scored 0.6× neural + 0.4× rule. The strongest 3 to 5 are shown.	`src/neurosymbolic/paths.py`
9. Reports	1,083 PDFs in five groups by recorded progression time (A to D, and N for none).	`scripts/build_patient_reports.py`

Every prediction is out-of-fold: five-fold cross-validation with seed 42, so no patient is scored by a model that saw them.

How the project evolved



Tried	What happened	Decision
Binary label on raw status	Censored patients counted as negatives, AUROC 0.54	Replaced by 36-month label and survival losses
Seven GNN architectures	None beat clinical logistic regression (AUROC 0.703)	Kept GraphSAGE as the neural part
Graph structure learning	First version hurt GCN (AUPRC 0.396 to 0.316), second within noise	Not adopted
SMOTE, GSE2034 graphs, RSF	No gain, and a linear model still won	Dropped
RuleKit IF-THEN rules	About 8 unweighted rules per fold, 10% using PrimeKG	Kept as baseline
MLN, PSL, RL, RL + MLN	Accuracy tied, but RL + MLN uses PrimeKG in 69% of formulas	RL + MLN is the engine

Results to remember

0.664
C-index of our model, tied with clinical Cox at 0.675

69%
of logic formulas use PrimeKG, against 10% for RuleKit

48%
of month estimates within 12 months, against 37% for a guess

Every model family stops near a C-index of 0.67. With 145 events, clinical facts carry most of the signal, so the neural and logic parts add explanation rather than accuracy.

Run it

```
pip install -r requirements.txt
python scripts/run_neurosymbolic_pipeline.py         # all five stages, about 50 min
python scripts/build_patient_reports.py --workers 4  # 1,083 reports, about 5 min
python scripts/export_demo_model.py                  # export the model for the demo
python demo/build_demo.py && node demo/check_engine.js
python -m unittest discover -s tests                 # 58 tests
```

Where to find things

Location	Contents
`outputs/patient_reports/`	Patient reports, group summaries, and the cohort overview
`notebooks/neurosymbolic_patient_reports.ipynb`	Walkthrough of the whole system
`notebooks/symbolic_reasoning_novelty.ipynb`	The logic reasoners and the novelty
`report/thesis.pdf`	Full project report, with all equations
`presentation/`	Panel deck, team guide, and this workflow
`demo/index.html`	Browser demo that runs the fitted model offline